LAB MODULE #4: 1/M REACTOR STARTUP LABORATORY

LABORATORY OBJECTIVES:

1. Manipulate control rods to increase the net reactivity in the PULSTAR reactor core and bring the reactor to criticality ($\rho_{net} = 0$) in a conservative manner.
2. Make conservative predictions for the critical rod height plotting 1/M factors as calculated from changes in the **Neutron Flux Monitor (NFM) Count Rate (CR)**.
3. Take the reactor critical and observe how well the **Actual Critical Position (ACP)** correlates with the **Projected Critical Position (PCP)**.

EQUIPMENT:

- PULSTAR Reactor.
- Neutron Flux Monitor (NFM).
- NFM Audio Channel.

PRECONDITIONS:

1. Reactor Daily Startup Checklist completed and signed by Designated Senior Reactor Operator (DSRO).
2. Reactor startup has been authorized by DSRO.
3. Licensed Reactor Operator is on duty at the reactor control console.
4. Verify that all control rods are placed 'ON GANG'.
5. Assure a conservative configuration for startup by verifying that the Startup Source (if used) is not positioned near the location of the NFM channel detector (a fission chamber near the PULSTAR core).

EQUATIONS:

$$\frac{1}{M} = \frac{CR_o}{CR_i}$$

Where:

M is the Subcritical Multiplication Factor.

CR_o is the initial Neutron Flux Monitor (NFM) Count Rate (CR) at startup.

CR_i is the NFM equilibrium count rate at each rod height.

PROCEDURE:

1. Record the initial NFM Count Rate (CR_o) on spreadsheet and observe auditory indication of the initial count rate on the NFM speaker.
2. Withdraw control rods ON GANG until either one of the following two conditions is met, *WHICHEVER HAPPENS FIRST*:
   a. The NFM Count Rate (CR) *DOUBLES*, **-or-**
   b. The control rods are withdrawn (starting from their current position) *HALF-WAY* to the Projected Critical Position (PCP) as given by the 1/M plot.

NOTE: There will not be a critical projection for the first rod pull, so criterial (2a) will be followed.

3. After subcritical equilibrium (i.e. a stable neutron population) has been obtained following each rod pull:
   a) Verify increased auditory count rate signal on NFM.
   b) Record current gang control rod height (*inches*) and NFM CR_i (*cps*) on the spreadsheet.
   c) Calculate a value for 1/M.
   d) Plot the 1/M value on the graph.
   e) Make a linear projection to the x-axis (criticality) with the Excel chart drawing tool using the two most recent 1/M points. The point where the linear 1/M fit crosses the x-axis is the PCP.
   f) Enter the PCP into the spreadsheet and determine the half-way point (HWP) between the current rod position and the PCP.
4. Repeat Step #2 until criticality is reached.

ANALYSIS & DISCUSSION:

1. Verify whether *ALL* of the 1/M projections were conservative. Discuss.
2. Discuss the significance of the non-linear relationship between control rod reactivity worth and control rod position.